

By submitting this assignment, I agree to the following:
 # "Aggies do not lie, cheat, or steal, or tolerate those who do."
 # "I have not given or received any unauthorized aid on this assignment."
 #
 # Names: Oliver Brumley
 # Iris Hernandez
 # Ruby Johnson
 # John Leach
 # Section: 559
 # Assignment: Lab 5 (Team)
 # Date: 09/14/2026

**no code, just test cases, due sunday 10pm
 **one submission on canvas for planning doc
 **everyone submit on gradescope for txt file (no header)

Champions' List:

Who	What	When
Everyone	Create Planning Doc	In Class
Ruby, Oliver	For the following variables: - sbp - med Ruby: Identify Normal, Corner, Edge cases. Write test cases. Oliver: Check test cases and output	Planning and test cases by 12pm Friday - 15 test cases Checking work by 5pm Sunday
Iris, Ruby	For the following variables: - hdl - sex - smo Iris: Identify Normal, Corner, Edge cases. Write test cases. Ruby: Check test cases and output	Planning and test cases by 12pm Friday - 15 test cases Checking work by 5pm Sunday
John, Iris	For the following variables: - cho John: Identify Normal, Corner, Edge cases. Write test cases. Iris: Check test cases and output	Planning and test cases by 12pm Friday - 15 test cases Checking work by 5pm Sunday

Oliver, John	For the following variables: - age Oliver: Identify Normal, Corner, Edge cases. Write test cases. John: Check test cases and output	Planning and test cases by 12pm Friday - 15 test cases Checking work by 5pm Sunday
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Example Test Case:

sex:F age:40 cho:105 smo:N hdl:60 sbp:100 med:N out:<1

Scoring System:

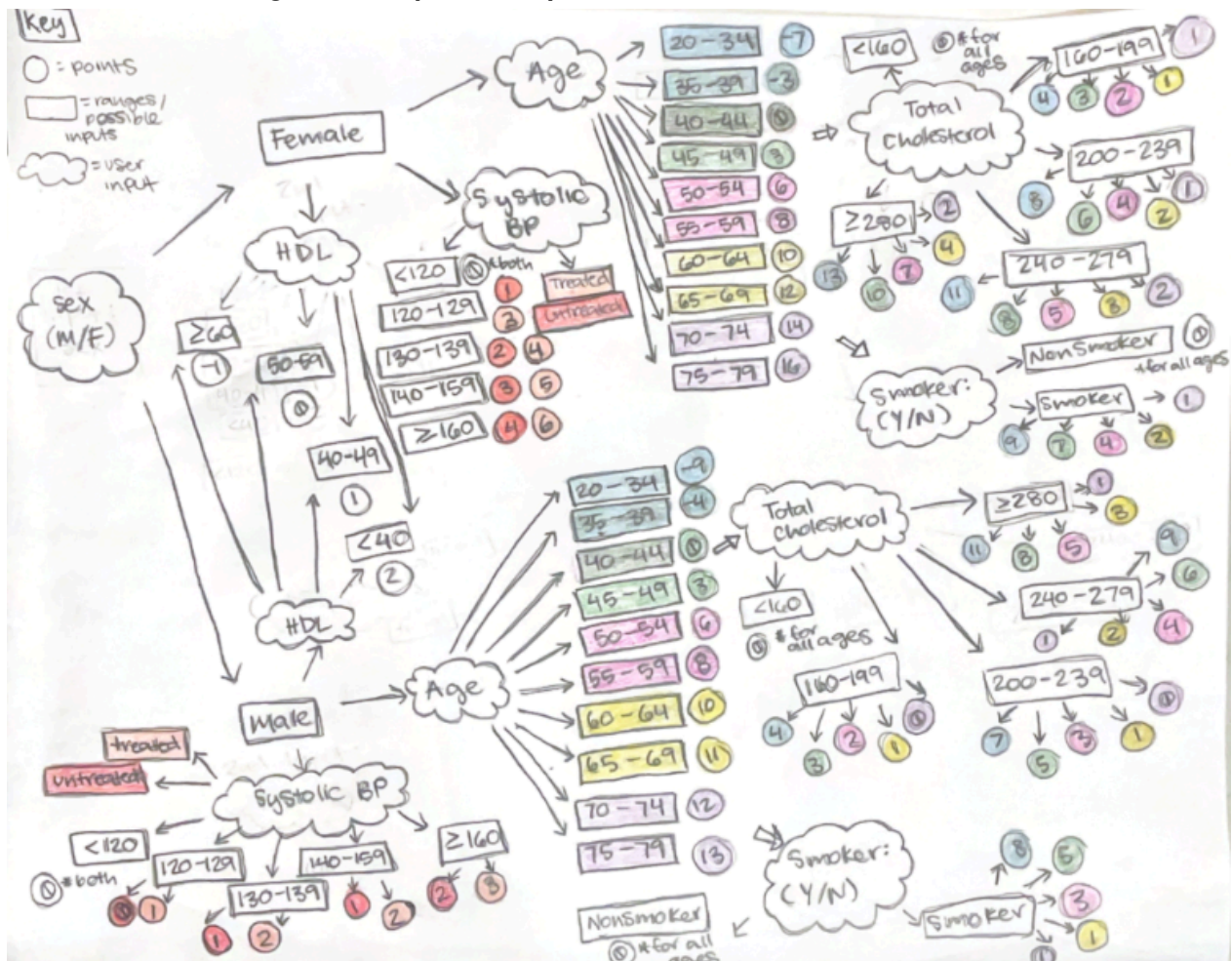
Variable				
Sex	Female		Male	
Age	Age		Age	
	Age	Points	Age	Points
	20-34	-7	20-34	-9
	35-39	-3	35-39	-4
	40-44	0	40-44	0
	45-49	3	45-49	3
	50-54	6	50-54	6
	55-59	8	55-59	8
	60-64	10	60-64	10
	65-69	12	65-69	11
	70-74	14	70-74	12

	<table><tr><td>75-79</td><td>16</td></tr></table>	75-79	16	<table><tr><td>75-79</td><td>13</td></tr></table>	75-79	13																																																																				
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Systolic BP and Medication Status	Systolic BP	<120	120-129	130-139	140-159	≥160
	Treated	0	3	4	5	6
	Untreated	0	1	2	3	4

Systolic BP and Medication Status	Systolic BP	<120	120-129	130-139	140-159	≥160
	Treated	0	1	2	2	3
	Untreated	0	0	1	1	2

Chart for determining intended points output:



Test Cases Planning:

General Test Case

sex:F age:40 cho:105 smo:N hdl:60 sbp:100 med:N out:<1

sex:M age:44 cho:158 smo:N hdl:51 sbp:110 med:N out:<1

Definitions of different cases used:

- Typical: Cases meant to ensure that the program is working generally. Often lie between parameters of variables being tested. Expected and standard.
- Edge: Cases containing values that test the absolute maximum or minimum of a single parameter.
- Corner: Cases which contain values which test the absolute maximum or minimum of multiple variables. Considered special cases.

Planning for all test cases for BP and Medication Status

Variable	Corner Case	Edge Case	Typical Case
Systolic BP (sbp)	Anything <120 - 106 - 115 - 100 - 129 - etc. Anything >= 160 - 160 - 161 - 180 - 195 - etc.	- 120 - 129 - 130 - 139 - 140 - 159	Anything between 120 and 129 - 121 - 125 - 128 - etc. Anything between 130 and 139 - 132 - 136 - 137 - Etc. Anything between 140 and 159 - 143 - 144 - 156 - etc.
Medication Status (med)	Y, N	none	none
Cases that cover each case	sbp:129 med:N sbp:160 med:Y sbp:160 med:N	sbp:120 med:Y sbp:129 med:N sbp:139 med:Y sbp:130 med:N sbp:140 med:Y sbp:159 med:N	sbp:121 med:Y sbp:128 med:N sbp:132 med:N sbp:137 med:Y sbp:144 med:Y sbp:156 med:N

Generic Cases for each test case with sbp and med changed to match cases needed above:

Corner Cases:

Women:

sex:F age:40 cho:105 smo:N hdl:50 sbp:129 med:N out:<1

sex:F age:41 cho:110 smo:N hdl:54 sbp:160 med:Y out:<1

sex:F age:42 cho:123 smo:N hdl:54 sbp:165 med:N out:<1

Men:

sex:M age:43 cho:137 smo:N hdl:50 sbp:129 med:N out:<1

sex:M age:44 cho:111 smo:N hdl:54 sbp:160 med:Y out:<1

sex:M age:40 cho:150 smo:N hdl:54 sbp:165 med:N out:<1

Edge Cases:

Women:

sex:F age:40 cho:105 smo:N hdl:50 sbp:120 med:Y out:<1

sex:F age:40 cho:105 smo:N hdl:50 sbp:129 med:N out:<1

sex:F age:40 cho:105 smo:N hdl:50 sbp:139 med:Y out:<1

sex:F age:40 cho:105 smo:N hdl:50 sbp:130 med:N out:<1

sex:F age:40 cho:105 smo:N hdl:50 sbp:140 med:Y out:<1

sex:F age:40 cho:105 smo:N hdl:50 sbp:159 med:Y out:<1

Men:

sex:M age:40 cho:105 smo:N hdl:50 sbp:120 med:Y out:<1

sex:M age:40 cho:105 smo:N hdl:50 sbp:129 med:N out:<1

sex:M age:40 cho:105 smo:N hdl:50 sbp:139 med:Y out:<1

sex:M age:40 cho:105 smo:N hdl:50 sbp:130 med:N out:<1

sex:M age:40 cho:105 smo:N hdl:50 sbp:140 med:Y out:<1

sex:M age:40 cho:105 smo:N hdl:50 sbp:159 med:N out:<1

Typical Cases:

Women:

sex:F age:40 cho:105 smo:N hdl:50 sbp:121 med:Y out:<1

sex:F age:40 cho:105 smo:N hdl:50 sbp:128 med:N out:<1

sex:F age:40 cho:105 smo:N hdl:50 sbp:132 med:N out:<1

sex:F age:40 cho:105 smo:N hdl:50 sbp:137 med:Y out:<1

sex:F age:40 cho:105 smo:N hdl:50 sbp:144 med:Y out:<1

sex:F age:40 cho:105 smo:N hdl:50 sbp:156 med:N out:<1

Men:

sex:M age:40 cho:105 smo:N hdl:50 sbp:121 med:Y out:<1

sex:M age:40 cho:105 smo:N hdl:50 sbp:128 med:N out:<1

sex:M age:40 cho:105 smo:N hdl:50 sbp:132 med:N out:<1

sex:M age:40 cho:105 smo:N hdl:50 sbp:137 med:Y out:<1

sex:M age:40 cho:105 smo:N hdl:50 sbp:144 med:Y out:<1

sex:M age:40 cho:105 smo:N hdl:50 sbp:156 med:N out:<1

Planning for all test cases for Age and Cholesterol

	Common Case	Edge case	Corner Case
Age	Between 20-34 21,30,33,... Between 35-39 36,37,... Between 40-44 41,48,... Between 45-49 46,47,... Between 50-54 53,52,... Between 55-59 56,57,58 Between 60-64 61,62,63 Between 65-69 66,67,68 Between 70-74 71,72,73 Between 75-79 76,77,78 sex:F age:52 cho:105 smo:N hdl:60 sbp:100 med:N out:<1	20,34,35,39,40,44,45 ,49,50,54,55,59,60,6 4,65,69,70,74,75,79 sex:F age:40 cho:105 smo:N hdl:60 sbp:100 med:N out:<1	Ages less than 20 yrs 19,18,17,... Ages over 79 yrs 80,81,82,... sex:F age:79 cho:105 smo:N hdl:60 sbp:100 med:N out:<1
Cholesterol	Between 160-199 161,180,179,... Between 200-239 221,230,... Between 240-279 242,257,263,270,... sex:F age:40 cho:105 smo:N hdl:60 sbp:100 med:N out:<1	160,199,200,239,240 ,279 sex:F age:40 cho:105 smo:N hdl:60 sbp:100 med:N out:<1	Less than 160 160,150,155,159,... Over 280 281,282,... sex:F age:40 cho:105 smo:N hdl:60 sbp:100 med:N out:<1

Planning for all test cases for HDL and Smoking Status

	Common	Edge	Corner
HDL Points dependent on sex	51,52,53,54,55,56,57,58 40,41,42,43,44,45,46,47,48	50,59,40,49	60,61,65,... 39,38,37,...
Smoker Dependent on your age and sex	No typical cases with Y/N answer	No edges with Y/N answer Male, 39yrs, smoker Male, 39yrs, nonsmoker Male, 40yrs, smoker Male, 40yrs, nonsmoker Male, 49yrs, smoker Male, 49yrs, nonsmoker Male, 50yrs, smoker Male, 50yrs, nonsmoker Male, 59yrs, smoker Male, 59yrs, nonsmoker Male, 60yrs, smoker Male, 60yrs, nonsmoker Male, 69yrs, smoker Male, 69yrs, nonsmoker Male, 70yrs, smoker Male, 70yrs, nonsmoker Male, 79yrs, smoker Male, 79yrs, nonsmoker And then again for females	Smoker, Nonsmoker Male, 20yrs, smoker Male, 20yrs, nonsmoker Female, 20yrs, smoker Female, 20yrs, nonsmoker Male, 79yrs, smoker Male, 79yrs, smoker Female, 79yrs, smoker Female, 79yrs, smoker
Sex	none	none	Female, Male

Outline to write all test cases:

- 1) Figure out base number of tests required accounting for Edge, Typical, and Corner cases
 - a) Without combining any tests, we need 118 cases to account for all calls in the tables above (an edge, corner, and typical case for each variable)
 - b) For all categories that rely on another answer, we will need double the test cases (ex: for hdl we need each test case for both female and male), which would mean we need about 228 total test cases
- 2) Combine test cases for each category - use one test case to account for required test of multiple variables to reduce number of tests we need to write

- a) If we combine the categories that fall under the same other answer (like how smoking status and systolic bp both are dependent on sex for point calculations) we can reduce the amount of cases we need
- 3) Put together all the test cases into single document in format of general test case above

Ruby - Blood Pressure & Medication Status

Planning for all test cases for BP and Medication Status

Variable	Corner Case	Edge Case	Typical Case
Systolic BP (sbp)	Anything <120 - 106 - 115 - 100 - 129 - etc. Anything >= 160 - 160 - 161 - 180 - 195 - etc.	- 120 - 129 - 130 - 139 - 140 - 159	Anything between 120 and 129 - 121 - 125 - 128 - etc. Anything between 130 and 139 - 132 - 136 - 137 - Etc. Anything between 140 and 159 - 143 - 144 - 156 - etc.
Medication Status (med)	Y, N	none	none
Cases that cover each case	sbp:129 med:N sbp:160 med:Y sbp:160 med:N	sbp:120 med:Y sbp:129 med:N sbp:139 med:Y sbp:130 med:N sbp:140 med:Y sbp:159 med:N	sbp:121 med:Y sbp:128 med:N sbp:132 med:N sbp:137 med:Y sbp:144 med:Y sbp:156 med:N

Generic Cases for each test case with sbp and med changed to match cases needed above:

Corner Cases:

Women:

sex:F age:40 cho:105 smo:N hdl:50 sbp:129 med:N out:<1

sex:F age:41 cho:110 smo:N hdl:54 sbp:160 med:Y out:<1

sex:F age:42 cho:123 smo:N hdl:54 sbp:165 med:N out:<1

Men:

sex:M age:43 cho:137 smo:N hdl:50 sbp:129 med:N out:<1

sex:M age:44 cho:111 smo:N hdl:54 sbp:160 med:Y out:<1

sex:M age:40 cho:150 smo:N hdl:54 sbp:165 med:N out:<1

Edge Cases:

Women:

sex:F age:40 cho:105 smo:N hdl:50 sbp:120 med:Y out:<1

sex:F age:40 cho:105 smo:N hdl:50 sbp:129 med:N out:<1

sex:F age:40 cho:105 smo:N hdl:50 sbp:139 med:Y out:<1

sex:F age:40 cho:105 smo:N hdl:50 sbp:130 med:N out:<1
sex:F age:40 cho:105 smo:N hdl:50 sbp:140 med:Y out:<1
sex:F age:40 cho:105 smo:N hdl:50 sbp:159 med:Y out:<1

Men:

sex:M age:40 cho:105 smo:N hdl:50 sbp:120 med:Y out:<1
sex:M age:40 cho:105 smo:N hdl:50 sbp:129 med:N out:<1
sex:M age:40 cho:105 smo:N hdl:50 sbp:139 med:Y out:<1
sex:M age:40 cho:105 smo:N hdl:50 sbp:130 med:N out:<1
sex:M age:40 cho:105 smo:N hdl:50 sbp:140 med:Y out:<1
sex:M age:40 cho:105 smo:N hdl:50 sbp:159 med:N out:<1

Typical Cases:

Women:

sex:F age:40 cho:105 smo:N hdl:50 sbp:121 med:Y out:<1
sex:F age:40 cho:105 smo:N hdl:50 sbp:128 med:N out:<1
sex:F age:40 cho:105 smo:N hdl:50 sbp:132 med:N out:<1
sex:F age:40 cho:105 smo:N hdl:50 sbp:137 med:Y out:<1
sex:F age:40 cho:105 smo:N hdl:50 sbp:144 med:Y out:<1
sex:F age:40 cho:105 smo:N hdl:50 sbp:156 med:N out:<1

Men:

sex:M age:40 cho:105 smo:N hdl:50 sbp:121 med:Y out:<1
sex:M age:40 cho:105 smo:N hdl:50 sbp:128 med:N out:<1
sex:M age:40 cho:105 smo:N hdl:50 sbp:132 med:N out:<1
sex:M age:40 cho:105 smo:N hdl:50 sbp:137 med:Y out:<1
sex:M age:40 cho:105 smo:N hdl:50 sbp:144 med:Y out:<1
sex:M age:40 cho:105 smo:N hdl:50 sbp:156 med:N out:<1

John - Age, Cholesterol

Planning for all test cases for Age and Cholesterol

	Common Case	Edge case	Corner Case
Age	sex:F age:52 cho:105 smo:N hdl:60 sbp:100 med:N out:<1	sex:F age:40 cho:105 smo:N hdl:60 sbp:100 med:N out:<1	sex:F age:79 cho:105 smo:N hdl:60 sbp:100 med:N out:<1
Cholesterol	sex:F age:40 cho:105 smo:N hdl:60 sbp:100 med:N out:<1	sex:F age:40 cho:105 smo:N hdl:60 sbp:100 med:N out:<1	sex:F age:40 cho:105 smo:N hdl:60 sbp:100 med:N out:<1

Iris - HDL, Sex, Smoker

Planning for all test cases for HDL and Smoking Status

	Common	Edge	Corner
HDL Dependent on sex and age	example(?) 52,48,29,66,24,26,4,25,64,32,45,73	50,59,40,49 Female, 50yrs Male, 50yrs, Female, 59yrs Male, 59yrs, Female, 59yrs	60, 39 Female, 39yrs Male, 39yrs Female, 60yrs Male, 60yrs
Smoker Dependent on your age and sex, and Y/N answer		Male, 39yrs, smoker Male, 39yrs, nonsmoker Male, 40yrs, smoker Male, 40yrs, nonsmoker Male, 49yrs, smoker Male, 49yrs, nonsmoker Male, 50yrs, smoker Male, 50yrs, nonsmoker Male, 59yrs, smoker Male, 59yrs, nonsmoker Male, 60yrs, smoker Male, 60yrs, nonsmoker Male, 69yrs, smoker Male, 69yrs, nonsmoker Male, 70yrs, smoker Male, 70yrs, nonsmoker Male, 79yrs, smoker Male, 79yrs, nonsmoker And then again for females	Male, 20yrs, smoker Male, 20yrs, nonsmoker Male, 79yrs, smoker (both equal 0 points, still be considered corner?) Male, 79yrs, smoker
Sex	none	none	Female, Male

Possible test cases

Example: sex:F age:40 cho:105 smo:N hdl:60 sbp:100 med:N out:<1

edge for smoker male

sex: M, age: 39, smo: N, hdl:60, sbp:100 med:N out:< (need to calculate, will exclude for now, and fill in variables)

Smoker corner

sex: M, age: 39, smo: Y, hdl:60, sbp:100 med:N out:<
sex: M, age: 39, smo: Y, hdl:60, sbp:100 med:N out:<
sex: M, age: 39, smo: N, hdl:60, sbp:100 med:N out:<
sex: F, age: 39, smo: Y, hdl:60, sbp:100 med:N out:<
sex: F, age: 39, smo: N, hdl:60, sbp:100 med:N out:<
sex: M, age: 79, smo: N, hdl:60, sbp:100 med:N out:<
sex: M, age: 79, smo: Y, hdl:60, sbp:100 med:N out:<
sex: F, age: 79, smo: N, hdl:60, sbp:100 med:N out:<
sex: F, age: 79, smo: Y, hdl:60, sbp:100 med:N out:<

Smoker edge

sex: M, age: 40, smo: Y, hdl:60, sbp:100 med:N out:<
sex: M, age: 49, smo: N, hdl:60, sbp:100 med:N out:<
sex: M, age: 49, smo: Y, hdl:60, sbp:100 med:N out:<
sex: M, age: 50, smo: N, hdl:60, sbp:100 med:N out:<
sex: M, age: 50, smo: Y, hdl:60, sbp:100 med:N out:<
sex: M, age: 59, smo: N, hdl:60, sbp:100 med:N out:<
sex: M, age: 59, smo: Y, hdl:60, sbp:100 med:N out:<
sex: M, age: 60, smo: N, hdl:60, sbp:100 med:N out:<
sex: M, age: 60, smo: Y, hdl:60, sbp:100 med:N out:<
sex: M, age: 69, smo: N, hdl:60, sbp:100 med:N out:<
sex: M, age: 69, smo: Y, hdl:60, sbp:100 med:N out:<
sex: M, age: 70, smo: N, hdl:60, sbp:100 med:N out:<
sex: M, age: 70, smo: Y, hdl:60, sbp:100 med:N out:<

sex: F, age: 40, smo: Y, hdl:60, sbp:100 med:N out:<
sex: F, age: 49, smo: N, hdl:60, sbp:100 med:N out:<
sex: F, age: 49, smo: Y, hdl:60, sbp:100 med:N out:<
sex: F, age: 50, smo: N, hdl:60, sbp:100 med:N out:<
sex: F, age: 50, smo: Y, hdl:60, sbp:100 med:N out:<
sex: F, age: 59, smo: N, hdl:60, sbp:100 med:N out:<
sex: F, age: 59, smo: Y, hdl:60, sbp:100 med:N out:<
sex: F, age: 60, smo: N, hdl:60, sbp:100 med:N out:<
sex: F, age: 60, smo: Y, hdl:60, sbp:100 med:N out:<
sex: F, age: 69, smo: N, hdl:60, sbp:100 med:N out:<

sex: F, age: 69, smo: Y, hdl:60, sbp:100 med:N out:<
sex: F, age: 70, smo: N, hdl:60, sbp:100 med:N out:<
sex: F, age: 70, smo: Y, hdl:60, sbp:100 med:N out:<

Hdl corner

sex: M, age: 39, smo: Y, hdl:60, sbp:100 med:N out:<

Hdl typical

H

Office Hour Notes

Tests cases in every planning document going forward

-Corner: values outside of ranges, invalid values (wont matter?), absolute maximum for the score

-Edge case will still work but at the edge

Corner case something that will not work at all

-typical : one or all variables within ranges

edge: all values are at the edge of their ranges

corner: values outside ranges

Shouldn't test them all, make it so that the program get more overworked

Typical edge and corner are examples

Define typical edge and corner in planning document

Write the formulas (?)

edge for smoker female

sex: F, age: 39, smo: N, hdl:60, sbp:100 med:N out:<

edge for hdl female

edge for hdl male

Oliver - Age, Sbp, Med



Figure out base number of tests required:

The number of unique parameter → point value mappings is equal to the sum of the number of point-value cells in the tables. The total number of point-value cells = 118 (unless I miscounted), so we need at least 118 test cases if we test each table independently (no combined tests yet).

Account for Edge, Middle, “Corner”

However, Shana wants us to account for edge, middle, and “corner” cases for each range. Her definition of “corner” case seems to be an edge case of a range that is itself on the edge of the table, so “corner” cases do not actually add to the total number of cases, because all corner cases are also edge cases ($\text{corner_cases} \subseteq \text{edge_cases}$).

This means that:

- For cells which take a range of inputs for one parameter, the number of test cases is multiplied by two, because for each output cell, we have to test:
 param_1: [range1, edge]
 param_1: [range1, middle]
- For cells which take a range of inputs for two parameters, the number of test cases is also multiplied by two, because these inputs map to the same cell:
 param_1: [range1, edge], param_2: [range1, edge]
 param_1: [range1, middle], param_2: [range1, middle]
 These combinations are redundant, and can be ignored:
 param_1: [range1, edge], param_2: [range1, middle]
 param_1: [range1, middle], param_2: [range1, edge]

Simple Combinatorics

So, if we test each cell *independently* (let's consider this simpler version first), the total sum of test cases is... (table identified with parameters, \$ indicates ranged value; a row with ≥ 1 \$ gets multiplied by (2)):

Age

$$\text{sex} * \$\text{age} = 2 * 10 * (2) = 40$$

Total Cholesterol

$$\text{sex} * \$\text{age} * \$\text{cho} = 2 * 5 * 5 * (2) = 100$$

HDL (mg/dl)

$$\$ \text{hdl} = 4 * (2) = 8$$

note: hdl does not include sex, because M/F hdl tables are identical

Smoking

$$\text{sex} * \$\text{age} * \text{smo} = 2 * 5 * 2 * (2) = 40$$

Systolic BP (mmHg)

$$\text{sex} * \$\text{spb} * \text{med} = 2 * 5 * 2 * (2) = 40$$

TOTAL TEST CASES

$$= 40 + 100 + 8 + 40 + 40 = 228$$

Optimization

This simpler method can be optimized, if we notice that the outputs for multiple tables can be tested in the same case.

(This is *technically* false because we are only testing the *sum* of all the tables' point outputs. If one table produces a value +1 point over the intended value, and another table produces a value -1 point *under* the intended value, the test case can still pass because the incorrect outputs will cancel in the sum-total. But let's assume for the sake of completing this assignment on time that the chance of this "error cancellation" is very slim. If I were designing this program, I would instead construct each table as its own function, and do unit-testing on each function rather than trying to test the entire algorithm at once— what was Shana saying about arches and pyramids again?)

So, we are changing the parameters for table_1 and table_2 in the same line. This will only work if table_1 and table_2 do not share any parameters, *or* if their parameter values conveniently coincide. Otherwise we could have a conflict, for example, if age for the Age table needs to change from range_1 (20-34) to range_2 (35-39) but for the Cholesterol table, it needs to change from range_1 (20-39) to range_2 (40-49). This is a fairly difficult problem.

Let's look at a case where the optimization is easy, because the tables have a sort of "parallel structure" with the same dimensions, same number of cells, and the only shared variable is sex:

Smoking

$$\text{sex} * \$\text{age} * \text{smo} = 2 * 5 * 2 * (2) = 40$$

Systolic BP (mmHg)

$$\text{sex} * \$\text{spb} * \text{med} = 2 * 5 * 2 * (2) = 40$$

Constructed as a loop to iterate over all combinations, the structure would look like:

Smoking

```
for sex:
  for age:
    (age = age_range)
    for smo:
      for case:
        (edge, middle)
        generate test
```

Systolic BP (mmHg)

```
for sex:
  for sbp:
    (sbp = sbp_range)
    for med:
```

```

for case:
    (edge, middle)
        generate test

```

These loops could be combined into something like:

Smoking and Systolic BP (mmHg)

```

for sex:
    for age: also change sbp
        (age = age_range), (sbp = sbp_range)
            for smo: also change med
                for case:
                    (edge, middle)
                        generate test

```

New total test cases = $228 - 40 = 188$

I don't want to do any more optimization, because the other tables do not fit together so nicely, and 188 is a reasonably small number (far under the number of test cases on an average leetcode problem). I'm going to use my findings here to write a Python script to generate these test cases and output them to a .txt file, because it's more fun than doing it by hand, and a more informative Python exercise than typing out all 188 tests (professional programmers aren't out there writing test cases by hand, I promise you!)

Point-Sum → Risk Conversion

The final consideration is that we also want to make sure to test that the point values are being converted into final risk percentages correctly. This is a separate testing problem, because it involves varying the total point-sum output from all the tables, rather than the range of outputs from each individual table. There is a chance that by independently testing the outputs from each table, we conveniently get the full range of sum-outputs. The easiest way I see to do this is by actually writing the tests to cover all the individual table outputs, and then filling in the gaps if we are still missing outputs in certain point-sum → risk ranges.

Initial Results + Tweaks

(Test-gen program and detailed explanation of how it works available upon request, because code is not allowed in the planning doc!) Basically, I copied the point-tables from the specification into n-D lists, wrote two simple functions to map input ranges to the list indices, another function to sum the point values, a third to map point sums to percent risks, and then I used loops analogous to the pseudocode example in "Optimization" to generate all the test cases, and write them to the .txt file. (The actual output string is generated by the `__str__` function in my `test_case` dataclass).

For my test-gen program, I ended up accounting for the lower corners separately, because it

simply made the loops nicer to check the upper edge for each range instead of trying to alternate (and landing on the upper corner for the final edge). Since my program was not as optimal as my plan, my initial set of generated cases was a bit larger, at 195.

After generating this initial set to cover each *table* output, my test coverage on gradescope was 94%, meaning I must be missing some *output-sum* ranges. I read through my generated cases and marked off each percent output (for each sex) when I saw an occurrence of it. After doing this, I sure enough found 8 missing outputs that were not covered by my base program:

```
# F Missing Output Values:
```

```
'''
```

```
[20, '11'],
```

```
[21, '14'],
```

```
[23, '22'],
```

```
[24, '27'],
```

```
[25, '≥30']
```

```
'''
```

```
# M Missing Output Values:
```

```
'''
```

```
[0, '1'],
```

```
[15, '20'],
```

```
[17, '≥30']
```

```
'''
```

I hardcoded the following 8 test cases to reach each of these output values (by tweaking generated cases that were only one point away):

```
sex:F age:75 cho:105 smo:N hdl:60 sbp:180 med:Y out:14
sex:F age:75 cho:105 smo:N hdl:40 sbp:180 med:Y out:22
sex:F age:75 cho:105 smo:N hdl:10 sbp:180 med:Y out:27
sex:F age:79 cho:300 smo:Y hdl:10 sbp:300 med:Y out:>30
sex:M age:35 cho:140 smo:N hdl:10 sbp:150 med:Y out:1
sex:M age:75 cho:105 smo:N hdl:10 sbp:100 med:N out:20
sex:M age:79 cho:300 smo:Y hdl:10 sbp:300 med:Y out:>30
```

That brought my total number of test cases to 203, and my test coverage to 100%.

Final Test Set:

```
sex:F age:20 cho:105 smo:N hdl:60 sbp:100 med:N out:<1
sex:F age:27 cho:105 smo:N hdl:60 sbp:100 med:N out:<1
sex:F age:34 cho:105 smo:N hdl:60 sbp:100 med:N out:<1
sex:F age:37 cho:105 smo:N hdl:60 sbp:100 med:N out:<1
sex:F age:39 cho:105 smo:N hdl:60 sbp:100 med:N out:<1
sex:F age:42 cho:105 smo:N hdl:60 sbp:100 med:N out:<1
```

sex:F age:44 cho:105 smo:N hdl:60 sbp:100 med:N out:<1
sex:F age:47 cho:105 smo:N hdl:60 sbp:100 med:N out:<1
sex:F age:49 cho:105 smo:N hdl:60 sbp:100 med:N out:<1
sex:F age:52 cho:105 smo:N hdl:60 sbp:100 med:N out:<1
sex:F age:54 cho:105 smo:N hdl:60 sbp:100 med:N out:<1
sex:F age:57 cho:105 smo:N hdl:60 sbp:100 med:N out:<1
sex:F age:59 cho:105 smo:N hdl:60 sbp:100 med:N out:<1
sex:F age:62 cho:105 smo:N hdl:60 sbp:100 med:N out:1
sex:F age:64 cho:105 smo:N hdl:60 sbp:100 med:N out:1
sex:F age:67 cho:105 smo:N hdl:60 sbp:100 med:N out:1
sex:F age:69 cho:105 smo:N hdl:60 sbp:100 med:N out:1
sex:F age:72 cho:105 smo:N hdl:60 sbp:100 med:N out:2
sex:F age:74 cho:105 smo:N hdl:60 sbp:100 med:N out:2
sex:F age:77 cho:105 smo:N hdl:60 sbp:100 med:N out:3
sex:F age:79 cho:105 smo:N hdl:60 sbp:100 med:N out:3
sex:M age:20 cho:105 smo:N hdl:60 sbp:100 med:N out:<1
sex:M age:27 cho:105 smo:N hdl:60 sbp:100 med:N out:<1
sex:M age:34 cho:105 smo:N hdl:60 sbp:100 med:N out:<1
sex:M age:37 cho:105 smo:N hdl:60 sbp:100 med:N out:<1
sex:M age:39 cho:105 smo:N hdl:60 sbp:100 med:N out:<1
sex:M age:42 cho:105 smo:N hdl:60 sbp:100 med:N out:<1
sex:M age:44 cho:105 smo:N hdl:60 sbp:100 med:N out:<1
sex:M age:47 cho:105 smo:N hdl:60 sbp:100 med:N out:1
sex:M age:49 cho:105 smo:N hdl:60 sbp:100 med:N out:1
sex:M age:52 cho:105 smo:N hdl:60 sbp:100 med:N out:2
sex:M age:54 cho:105 smo:N hdl:60 sbp:100 med:N out:2
sex:M age:57 cho:105 smo:N hdl:60 sbp:100 med:N out:3
sex:M age:59 cho:105 smo:N hdl:60 sbp:100 med:N out:3
sex:M age:62 cho:105 smo:N hdl:60 sbp:100 med:N out:5
sex:M age:64 cho:105 smo:N hdl:60 sbp:100 med:N out:5
sex:M age:67 cho:105 smo:N hdl:60 sbp:100 med:N out:6
sex:M age:69 cho:105 smo:N hdl:60 sbp:100 med:N out:6
sex:M age:72 cho:105 smo:N hdl:60 sbp:100 med:N out:8
sex:M age:74 cho:105 smo:N hdl:60 sbp:100 med:N out:8
sex:M age:77 cho:105 smo:N hdl:60 sbp:100 med:N out:10
sex:M age:79 cho:105 smo:N hdl:60 sbp:100 med:N out:10
sex:F age:20 cho:0 smo:N hdl:60 sbp:100 med:N out:<1
sex:F age:30 cho:120 smo:N hdl:60 sbp:100 med:N out:<1

sex:F age:39 cho:159 smo:N hdl:60 sbp:100 med:N out:<1
sex:F age:45 cho:120 smo:N hdl:60 sbp:100 med:N out:<1
sex:F age:49 cho:159 smo:N hdl:60 sbp:100 med:N out:<1
sex:F age:55 cho:120 smo:N hdl:60 sbp:100 med:N out:<1
sex:F age:59 cho:159 smo:N hdl:60 sbp:100 med:N out:<1
sex:F age:65 cho:120 smo:N hdl:60 sbp:100 med:N out:1
sex:F age:69 cho:159 smo:N hdl:60 sbp:100 med:N out:1
sex:F age:75 cho:120 smo:N hdl:60 sbp:100 med:N out:3
sex:F age:79 cho:159 smo:N hdl:60 sbp:100 med:N out:3
sex:F age:30 cho:180 smo:N hdl:60 sbp:100 med:N out:<1
sex:F age:39 cho:199 smo:N hdl:60 sbp:100 med:N out:<1
sex:F age:45 cho:180 smo:N hdl:60 sbp:100 med:N out:<1
sex:F age:49 cho:199 smo:N hdl:60 sbp:100 med:N out:<1
sex:F age:55 cho:180 smo:N hdl:60 sbp:100 med:N out:1
sex:F age:59 cho:199 smo:N hdl:60 sbp:100 med:N out:1
sex:F age:65 cho:180 smo:N hdl:60 sbp:100 med:N out:1
sex:F age:69 cho:199 smo:N hdl:60 sbp:100 med:N out:1
sex:F age:75 cho:180 smo:N hdl:60 sbp:100 med:N out:4
sex:F age:79 cho:199 smo:N hdl:60 sbp:100 med:N out:4
sex:F age:30 cho:220 smo:N hdl:60 sbp:100 med:N out:<1
sex:F age:39 cho:239 smo:N hdl:60 sbp:100 med:N out:<1
sex:F age:45 cho:220 smo:N hdl:60 sbp:100 med:N out:<1
sex:F age:49 cho:239 smo:N hdl:60 sbp:100 med:N out:<1
sex:F age:55 cho:220 smo:N hdl:60 sbp:100 med:N out:1
sex:F age:59 cho:239 smo:N hdl:60 sbp:100 med:N out:1
sex:F age:65 cho:220 smo:N hdl:60 sbp:100 med:N out:2
sex:F age:69 cho:239 smo:N hdl:60 sbp:100 med:N out:2
sex:F age:75 cho:220 smo:N hdl:60 sbp:100 med:N out:4
sex:F age:79 cho:239 smo:N hdl:60 sbp:100 med:N out:4
sex:F age:30 cho:260 smo:N hdl:60 sbp:100 med:N out:<1
sex:F age:39 cho:279 smo:N hdl:60 sbp:100 med:N out:<1
sex:F age:45 cho:260 smo:N hdl:60 sbp:100 med:N out:1
sex:F age:49 cho:279 smo:N hdl:60 sbp:100 med:N out:1
sex:F age:55 cho:260 smo:N hdl:60 sbp:100 med:N out:1
sex:F age:59 cho:279 smo:N hdl:60 sbp:100 med:N out:1
sex:F age:65 cho:260 smo:N hdl:60 sbp:100 med:N out:2
sex:F age:69 cho:279 smo:N hdl:60 sbp:100 med:N out:2
sex:F age:75 cho:260 smo:N hdl:60 sbp:100 med:N out:5

sex:F age:79 cho:279 smo:N hdl:60 sbp:100 med:N out:5
sex:F age:30 cho:300 smo:N hdl:60 sbp:100 med:N out:<1
sex:F age:39 cho:999 smo:N hdl:60 sbp:100 med:N out:1
sex:F age:45 cho:300 smo:N hdl:60 sbp:100 med:N out:1
sex:F age:49 cho:999 smo:N hdl:60 sbp:100 med:N out:1
sex:F age:55 cho:300 smo:N hdl:60 sbp:100 med:N out:2
sex:F age:59 cho:999 smo:N hdl:60 sbp:100 med:N out:2
sex:F age:65 cho:300 smo:N hdl:60 sbp:100 med:N out:3
sex:F age:69 cho:999 smo:N hdl:60 sbp:100 med:N out:3
sex:F age:75 cho:300 smo:N hdl:60 sbp:100 med:N out:5
sex:F age:79 cho:999 smo:N hdl:60 sbp:100 med:N out:5
sex:M age:20 cho:0 smo:N hdl:60 sbp:100 med:N out:<1
sex:M age:30 cho:120 smo:N hdl:60 sbp:100 med:N out:<1
sex:M age:39 cho:159 smo:N hdl:60 sbp:100 med:N out:<1
sex:M age:45 cho:120 smo:N hdl:60 sbp:100 med:N out:1
sex:M age:49 cho:159 smo:N hdl:60 sbp:100 med:N out:1
sex:M age:55 cho:120 smo:N hdl:60 sbp:100 med:N out:3
sex:M age:59 cho:159 smo:N hdl:60 sbp:100 med:N out:3
sex:M age:65 cho:120 smo:N hdl:60 sbp:100 med:N out:6
sex:M age:69 cho:159 smo:N hdl:60 sbp:100 med:N out:6
sex:M age:75 cho:120 smo:N hdl:60 sbp:100 med:N out:10
sex:M age:79 cho:159 smo:N hdl:60 sbp:100 med:N out:10
sex:M age:30 cho:180 smo:N hdl:60 sbp:100 med:N out:<1
sex:M age:39 cho:199 smo:N hdl:60 sbp:100 med:N out:<1
sex:M age:45 cho:180 smo:N hdl:60 sbp:100 med:N out:2
sex:M age:49 cho:199 smo:N hdl:60 sbp:100 med:N out:2
sex:M age:55 cho:180 smo:N hdl:60 sbp:100 med:N out:5
sex:M age:59 cho:199 smo:N hdl:60 sbp:100 med:N out:5
sex:M age:65 cho:180 smo:N hdl:60 sbp:100 med:N out:8
sex:M age:69 cho:199 smo:N hdl:60 sbp:100 med:N out:8
sex:M age:75 cho:180 smo:N hdl:60 sbp:100 med:N out:10
sex:M age:79 cho:199 smo:N hdl:60 sbp:100 med:N out:10
sex:M age:30 cho:220 smo:N hdl:60 sbp:100 med:N out:<1
sex:M age:39 cho:239 smo:N hdl:60 sbp:100 med:N out:1
sex:M age:45 cho:220 smo:N hdl:60 sbp:100 med:N out:3
sex:M age:49 cho:239 smo:N hdl:60 sbp:100 med:N out:3
sex:M age:55 cho:220 smo:N hdl:60 sbp:100 med:N out:6
sex:M age:59 cho:239 smo:N hdl:60 sbp:100 med:N out:6

sex:M age:65 cho:220 smo:N hdl:60 sbp:100 med:N out:8
sex:M age:69 cho:239 smo:N hdl:60 sbp:100 med:N out:8
sex:M age:75 cho:220 smo:N hdl:60 sbp:100 med:N out:10
sex:M age:79 cho:239 smo:N hdl:60 sbp:100 med:N out:10
sex:M age:30 cho:260 smo:N hdl:60 sbp:100 med:N out:<1
sex:M age:39 cho:279 smo:N hdl:60 sbp:100 med:N out:1
sex:M age:45 cho:260 smo:N hdl:60 sbp:100 med:N out:4
sex:M age:49 cho:279 smo:N hdl:60 sbp:100 med:N out:4
sex:M age:55 cho:260 smo:N hdl:60 sbp:100 med:N out:8
sex:M age:59 cho:279 smo:N hdl:60 sbp:100 med:N out:8
sex:M age:65 cho:260 smo:N hdl:60 sbp:100 med:N out:10
sex:M age:69 cho:279 smo:N hdl:60 sbp:100 med:N out:10
sex:M age:75 cho:260 smo:N hdl:60 sbp:100 med:N out:12
sex:M age:79 cho:279 smo:N hdl:60 sbp:100 med:N out:12
sex:M age:30 cho:300 smo:N hdl:60 sbp:100 med:N out:1
sex:M age:39 cho:999 smo:N hdl:60 sbp:100 med:N out:2
sex:M age:45 cho:300 smo:N hdl:60 sbp:100 med:N out:6
sex:M age:49 cho:999 smo:N hdl:60 sbp:100 med:N out:6
sex:M age:55 cho:300 smo:N hdl:60 sbp:100 med:N out:10
sex:M age:59 cho:999 smo:N hdl:60 sbp:100 med:N out:10
sex:M age:65 cho:300 smo:N hdl:60 sbp:100 med:N out:12
sex:M age:69 cho:999 smo:N hdl:60 sbp:100 med:N out:12
sex:M age:75 cho:300 smo:N hdl:60 sbp:100 med:N out:12
sex:M age:79 cho:999 smo:N hdl:60 sbp:100 med:N out:12
sex:F age:40 cho:105 smo:N hdl:60 sbp:100 med:N out:<1
sex:F age:40 cho:105 smo:N hdl:70 sbp:100 med:N out:<1
sex:F age:40 cho:105 smo:N hdl:999 sbp:100 med:N out:<1
sex:F age:40 cho:105 smo:N hdl:55 sbp:100 med:N out:<1
sex:F age:40 cho:105 smo:N hdl:59 sbp:100 med:N out:<1
sex:F age:40 cho:105 smo:N hdl:45 sbp:100 med:N out:<1
sex:F age:40 cho:105 smo:N hdl:49 sbp:100 med:N out:<1
sex:F age:40 cho:105 smo:N hdl:30 sbp:100 med:N out:<1
sex:F age:40 cho:105 smo:N hdl:39 sbp:100 med:N out:<1
sex:F age:20 cho:105 smo:N hdl:60 sbp:0 med:N out:<1
sex:F age:30 cho:105 smo:N hdl:60 sbp:100 med:N out:<1
sex:F age:39 cho:105 smo:N hdl:60 sbp:119 med:N out:<1
sex:F age:45 cho:105 smo:N hdl:60 sbp:125 med:N out:<1
sex:F age:49 cho:105 smo:N hdl:60 sbp:129 med:N out:<1

sex:F age:55 cho:105 smo:N hdl:60 sbp:135 med:N out:1
sex:F age:59 cho:105 smo:N hdl:60 sbp:139 med:N out:1
sex:F age:65 cho:105 smo:N hdl:60 sbp:150 med:N out:2
sex:F age:69 cho:105 smo:N hdl:60 sbp:159 med:N out:2
sex:F age:75 cho:105 smo:N hdl:60 sbp:180 med:N out:8
sex:F age:79 cho:105 smo:N hdl:60 sbp:999 med:N out:8
sex:F age:30 cho:105 smo:Y hdl:60 sbp:100 med:Y out:<1
sex:F age:39 cho:105 smo:Y hdl:60 sbp:119 med:Y out:<1
sex:F age:45 cho:105 smo:Y hdl:60 sbp:125 med:Y out:1
sex:F age:49 cho:105 smo:Y hdl:60 sbp:129 med:Y out:1
sex:F age:55 cho:105 smo:Y hdl:60 sbp:135 med:Y out:3
sex:F age:59 cho:105 smo:Y hdl:60 sbp:139 med:Y out:3
sex:F age:65 cho:105 smo:Y hdl:60 sbp:150 med:Y out:6
sex:F age:69 cho:105 smo:Y hdl:60 sbp:159 med:Y out:6
sex:F age:75 cho:105 smo:Y hdl:60 sbp:180 med:Y out:17
sex:F age:79 cho:105 smo:Y hdl:60 sbp:999 med:Y out:17
sex:M age:20 cho:105 smo:Y hdl:60 sbp:0 med:Y out:<1
sex:M age:30 cho:105 smo:N hdl:60 sbp:100 med:N out:<1
sex:M age:39 cho:105 smo:N hdl:60 sbp:119 med:N out:<1
sex:M age:45 cho:105 smo:N hdl:60 sbp:125 med:N out:1
sex:M age:49 cho:105 smo:N hdl:60 sbp:129 med:N out:1
sex:M age:55 cho:105 smo:N hdl:60 sbp:135 med:N out:4
sex:M age:59 cho:105 smo:N hdl:60 sbp:139 med:N out:4
sex:M age:65 cho:105 smo:N hdl:60 sbp:150 med:N out:8
sex:M age:69 cho:105 smo:N hdl:60 sbp:159 med:N out:8
sex:M age:75 cho:105 smo:N hdl:60 sbp:180 med:N out:16
sex:M age:79 cho:105 smo:N hdl:60 sbp:999 med:N out:16
sex:M age:30 cho:105 smo:Y hdl:60 sbp:100 med:Y out:<1
sex:M age:39 cho:105 smo:Y hdl:60 sbp:119 med:Y out:1
sex:M age:45 cho:105 smo:Y hdl:60 sbp:125 med:Y out:4
sex:M age:49 cho:105 smo:Y hdl:60 sbp:129 med:Y out:4
sex:M age:55 cho:105 smo:Y hdl:60 sbp:135 med:Y out:10
sex:M age:59 cho:105 smo:Y hdl:60 sbp:139 med:Y out:10
sex:M age:65 cho:105 smo:Y hdl:60 sbp:150 med:Y out:12
sex:M age:69 cho:105 smo:Y hdl:60 sbp:159 med:Y out:12
sex:M age:75 cho:105 smo:Y hdl:60 sbp:180 med:Y out:25
sex:M age:79 cho:105 smo:Y hdl:60 sbp:999 med:Y out:25
sex:F age:75 cho:105 smo:Y hdl:60 sbp:180 med:N out:11

sex:F age:75 cho:105 smo:N hdl:60 sbp:180 med:Y out:14
sex:F age:75 cho:105 smo:N hdl:40 sbp:180 med:Y out:22
sex:F age:75 cho:105 smo:N hdl:10 sbp:180 med:Y out:27
sex:F age:79 cho:300 smo:Y hdl:10 sbp:300 med:Y out:>30
sex:M age:35 cho:140 smo:N hdl:10 sbp:150 med:Y out:1
sex:M age:75 cho:105 smo:N hdl:10 sbp:100 med:N out:20
sex:M age:79 cho:300 smo:Y hdl:10 sbp:300 med:Y out:>30